

Imagine-Then-Act Chunk Selection for Lightweight Vision-Language-Action Policies

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Abstract—Lightweight vision-language-action (VLA) policies make language-conditioned robot control more accessible, but a single predicted action chunk can commit the robot to a poor local choice before any semantic feedback is available. We study a test-time wrapper for frozen lightweight VLA policies that separates three mechanisms: proposing diverse action-chunk candidates, selecting among them before execution, and measuring final task success with the environment. The wrapper generates a portfolio of subgoal- and prompt-conditioned candidate chunks, executes them only through the frozen VLA policy, and evaluates pre-execution choices through a simulation- or world-model-like selection module. In an initial candidate-generation study with SmolVLA, native policy noise reached real action-chunk inference but produced limited diversity (0.048 mean normalized pairwise L_2 , 0.0012 mean cosine distance), whereas a template prompt portfolio produced substantially more diverse policy-generated chunks (0.226 mean normalized pairwise L_2 , 0.030 mean cosine distance). We present this as evidence that wrapper-level proposal generation is a necessary first step toward imagined chunk selection, not as a claim of benchmark success improvement. The paper frames a conservative architecture for improving frozen lightweight VLA policies while keeping the success criterion external: final success is always the benchmark or environment signal, never the VLM or simulator selector itself.

I. INTRODUCTION

Vision-language-action models have expanded the scope of language-conditioned robot control by mapping images and instructions directly to robot actions. Large VLA systems such as RT-2, OpenVLA, Octo, and π_0 connect broad visual-language representations to manipulation policies [20, 11, 14, 2]. At the same time, practical deployment on low-cost arms, consumer GPUs, and lab-scale iteration loops motivates smaller policies such as SmolVLA [17]. These lightweight policies are attractive because they reduce the cost of experimentation and deployment, yet their compactness also makes test-time robustness especially important: a single action chunk may encode an ambiguous grasp, a premature placement, or a trajectory that does not match the user’s intended subgoal.

The usual path to improving a VLA policy is to collect more data or fine-tune the model. This work studies a complementary path: keep the lightweight VLA frozen, and add an external test-time layer that makes the policy’s action choices more inspectable before committing them to the environment. The core idea is simple. Instead of asking a frozen VLA for one action chunk, an external module proposes a small portfolio of subgoal or prompt variants. The same frozen VLA then produces a candidate action chunk for each variant. A selector evaluates the imagined consequence of each candidate, using a simulator clone, a learned visual dynamics model,

or a vision-language judge, and commits only the selected chunk to the actual environment. Final task success remains the benchmark success flag.

This framing is intentionally different from using a VLM as a success detector or from claiming a new model-predictive control algorithm. Our contribution is to organize a wrapper around a frozen lightweight VLA so that candidate proposal, candidate selection, and benchmark success remain separate quantities. This separation matters empirically. A naive best-of- N system requires candidate diversity, but an initial study showed that native SmolVLA noise produced action chunks that were still very similar. In contrast, a template prompt portfolio produced meaningfully larger action-chunk separation while still routing every candidate through the real SmolVLA policy. This suggests that wrapper-level proposal generation, rather than native policy noise alone, is the appropriate object of study for test-time chunk selection.

Figure 1 summarizes the proposed architecture. A portfolio generator expands an ambiguous task into candidate subgoals or prompts; the frozen VLA executor produces candidate chunks; a simulation- or world-model-like module predicts or checks candidate outcomes; and a selector chooses a chunk for execution. The loop can be repeated when the benchmark state indicates partial progress or failure, but the evaluation criterion is unchanged: only environment success is reported as task success.

The resulting paper asks a bounded question: can an external wrapper improve the usefulness of frozen lightweight VLA action proposals before any fine-tuning or model replacement? We make three contributions toward that question. First, we formulate imagined action-chunk selection as a modular wrapper around frozen lightweight VLA execution. Second, we show that subgoal/prompt portfolios can induce substantially more diverse SmolVLA action chunks than native noise. Third, we define an evaluation protocol that separates candidate diversity, selector quality, and environment-measured benchmark success.

II. RELATED WORK

A. Visual Foresight and Model-Predictive Selection

The idea of evaluating possible futures before execution is central to visual foresight and model-predictive control. Visual foresight methods learn predictive visual models and use them to select actions whose predicted images move toward a goal [6]. PETS uses probabilistic ensembles and trajectory sampling for model-based reinforcement learning [5], while CEM and MPPI-style procedures generate candidate control sequences

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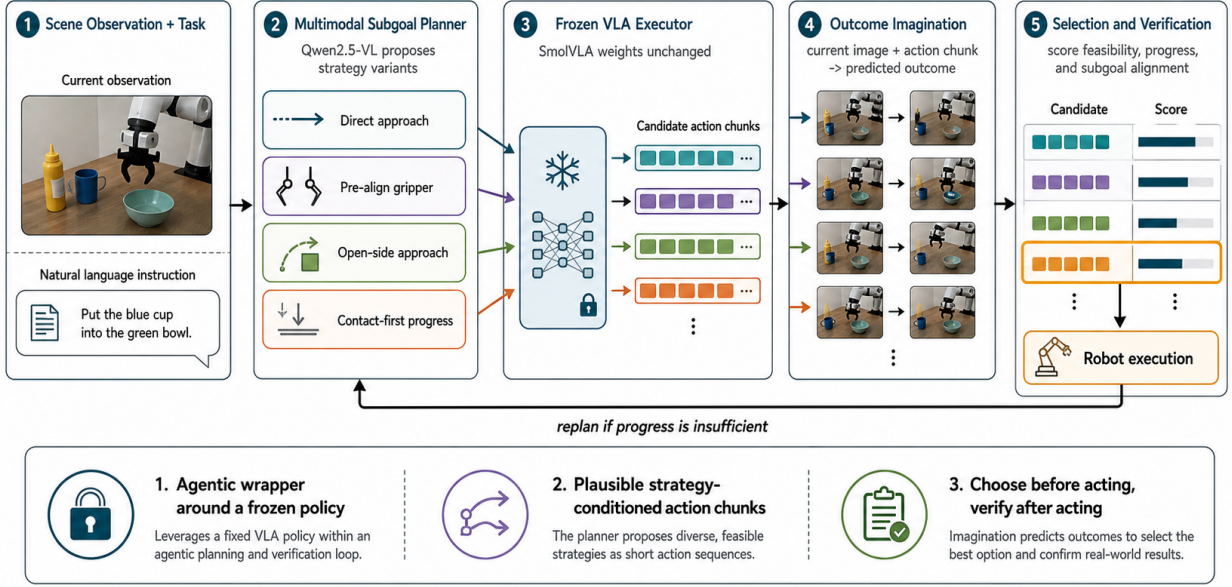


Fig. 1. Overview of Imagine-Then-Act. An external multimodal planner proposes subgoal-conditioned strategy variants, a frozen lightweight VLA produces candidate action chunks, an action-conditioned imagination module predicts or checks candidate outcomes, and a selection/verification loop executes the chosen chunk and replans when progress is insufficient. Final success remains the benchmark environment success signal.

and rank them by a model or cost. Recent work such as VLMPC and Prompting with the Future connects model-predictive ideas to vision-language evaluation and digital-twin rollouts [19, 13]. Our method is not a claim of novelty over this line. Instead, it asks how the same candidate-selection principle can be wrapped around frozen lightweight VLA action chunks, where the low-level action generator is a pretrained VLA rather than a hand-designed action sampler.

B. Multimodal Action Generation

For candidate selection to matter, the system must first generate distinct candidate behaviors. Diffusion Policy and Diffuser produce diverse trajectories through stochastic denoising [4, 10]. ACT uses action chunking for imitation learning on low-cost robot hardware [18]. Behavior Transformer and Implicit Behavioral Cloning model multimodal behavior distributions through learned action modes or implicit energy functions [16, 7]. These methods build diversity into the policy or planner itself. In contrast, we study a frozen VLA and create diversity externally, through subgoal and prompt portfolios that still require the VLA to generate every executable action chunk.

C. Language and Vision-Language Reasoning for Robotics

Language-model and vision-language systems often improve robot behavior by reasoning over skills, subgoals, or affordances rather than directly producing low-level motor commands. SayCan grounds language-model plans in learned

skill affordances [1]; Inner Monologue incorporates environmental feedback into language-conditioned planning [8]; Code as Policies converts language instructions into executable robot programs [12]; and VoxPoser composes 3D value maps for language-guided manipulation [9]. Our wrapper uses a similar separation between semantic reasoning and low-level control, but the interface is an action-chunk portfolio for a frozen VLA. A VLM may generate candidate subgoals or score imagined outcomes, but it is not the robot policy and does not define task success.

D. Vision-Language-Action Policies

VLA policies connect language-conditioned perception to robot action. RT-1 scaled transformer policies on robot data [3], and RT-2 showed that vision-language models can transfer web knowledge to robotic action prediction [20]. OpenVLA released an open-source VLA trained on a large mixture of robot demonstrations [11], and Octo provided an open generalist policy across embodiments [14]. Recent systems such as π_0 and FAST explore continuous flow-based action generation and efficient action tokenization for high-rate robot control [2, 15]. SmolVLA pushes this direction toward smaller and more accessible models, with a 450M-parameter architecture, asynchronous inference, and action chunking [17]. This work keeps such a lightweight VLA frozen and studies how a test-time wrapper can structure its candidate action chunks before execution.

TABLE I

EVALUATION AXES FOR THE IMAGINE-THEN-ACT WRAPPER. EACH AXIS IS REPORTED WITH ITS OWN METRIC, AND ONLY ENVIRONMENT-DEFINED COMPLETION IS TREATED AS TASK SUCCESS.

Axis	Measures	Success source
Candidate diversity	Chunk separation	Action metrics
Selector quality	Pre-execution ranking	Sim/proxy checks
Benchmark success	Task completion	Environment flag

III. METHOD

The proposed wrapper has four modules. The portfolio generator receives the current observation and task instruction and produces candidate subgoals or prompt variants. The frozen VLA executor maps each candidate prompt to an action chunk. The imagination module predicts or checks the consequence of each chunk, using either a simulator clone, a learned visual dynamics model, or a visual judge over predicted outcome images. The selector chooses one chunk for execution in the benchmark environment.

The architecture deliberately exposes two interfaces. The proposal interface controls how candidate chunks are generated. This can include template prompt portfolios, external VLM-generated subgoals, or structured perturbations around the nominal VLA chunk. The selection interface controls how candidate chunks are ranked before commitment. This can include oracle simulator rollouts for upper-bound analysis, non-oracle proxy checks, or image-based VLM scoring. In all cases, the committed action is still produced by the frozen VLA policy.

IV. EVALUATION PROTOCOL

The evaluation separates three outcomes. Candidate diversity measures whether the proposal interface creates meaningfully different action chunks. Selector quality measures whether the pre-execution selector chooses candidates that lead to better predicted or simulated outcomes than random selection. Benchmark success measures whether the committed policy execution solves the task according to the environment success condition. Only the third quantity is reported as task success.

V. LIMITATIONS

The current formulation adds test-time computation by evaluating multiple candidate chunks before execution. Simulator-based selection is also an upper-bound or proxy mechanism unless replaced by a deployable learned dynamics model. Prompt portfolios can increase action diversity, but diversity alone does not imply better selection or higher success. Finally, because the base VLA is frozen, the wrapper can only steer among behaviors that the policy can already express under some prompt or subgoal context.

VI. CONCLUSION

Imagine-then-act chunk selection provides a structured way to study test-time agentic wrappers for lightweight VLA policies. By separating proposal generation, pre-execution

selection, and environment-measured success, the framework avoids treating an auxiliary VLM or simulator score as task success. Initial candidate-generation evidence suggests that prompt portfolios can produce more diverse frozen-policy action chunks than native policy noise, motivating selection mechanisms that evaluate imagined outcomes before committing actions to the environment.

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